

INHERENT UNCERTAINTY OF HYPERBOLIC EMBEDDINGS OF COMPLEX NETWORKS

NETSci 2023 — NETWORK GEOMETRY (13B)

Ariane Lizotte, Jean-Gabriel Young and Antoine Allard

July 12, 2023

ariane.lizotte.2@ulaval.ca

Fonds de recherche
Nature et
technologies



Sentinel
North



Digital Research
Alliance of Canada



UNIVERSITÉ
LAVAL



Dynamica

Hyperbolic space: a natural network geometry

Networks obtained from hyperbolic geometry have properties that *match empirical observations*:

- degree sequence;
- *small-worldness*;
- shortest paths;
- community structure.

ARTICLES

PUBLISHED ONLINE: 16 NOVEMBER 2008 | DOI: 10.1038/NPHYS1130

nature
physics

Navigability of complex networks

Marián Boguñá^{1*}, Dmitri Krioukov² and K. C. Claffy²

scientific reports

Check for updates

OPEN

The inherent community structure of hyperbolic networks

Bianka Kovács¹ & Gergely Palla^{1,2,3,4,5}

REVIEWS

Check for updates

Network geometry

Marián Boguñá^{1,2}, Ivan Bonamassa⁴, Mantio De Domenico^{4,5}, Shlomo Havlin⁴, Dmitri Krioukov^{1,6,7,8} and M. Angeles Serrano^{1,2,9}

Abstract | Networks can be characterized as being hyperbolic, with distances defined by the shortest paths

PHYSICAL REVIEW RESEARCH 2, 043113 (2020)

Link prediction with hyperbolic geometry

Maksim Kitsak^{1,2}, Ivan Voitalov^{3,2} and Dmitri Krioukov^{1,2}

¹Faculty of Electrical Engineering, Delft University of Technology; Mathematics and Computer Science, 2600 GA Delft, The Netherlands

Hyperbolic space: a natural network geometry

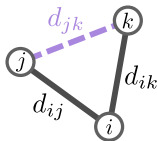
Networks obtained from hyperbolic geometry have properties that *match empirical observations*:

- degree sequence;
- *small-worldness*;
- shortest paths;
- community structure.

Chiefly, the triangle inequality

$$d_{jk} \leq d_{ij} + d_{ik}$$

naturally induces clustering.



ARTICLES

PUBLISHED ONLINE: 16 NOVEMBER 2008 | DOI: 10.1038/NPHYS1130

nature
physics

Navigability of complex networks

Marián Boguñá^{1*}, Dmitri Krioukov² and K. C. Claffy²

scientific reports

OPEN

Check for updates

The inherent community structure of hyperbolic networks

Bianka Kovács¹ & Gergely Palla^{1,2,3,†}

REVIEWS

Check for updates

Network geometry

Marián Boguñá^{1,2}, Ivan Bonamassa⁴, Mantio De Domenico^{4,†}, Shlomo Havlin⁴, Dmitri Krioukov^{5,6,7,8} and M. Angeles Serrano^{1,2,9}

Abstract | Networks can be characterized using a distance defined by the shortest paths

PHYSICAL REVIEW RESEARCH 2, 043113 (2020)

Link prediction with hyperbolic geometry

Maksim Kitsak^{1,2}, Ivan Voitalov^{3,2} and Dmitri Krioukov^{4,2}

¹Faculty of Electrical Engineering, Delft University of Technology; Mathematics and Computer Science, 2600 GA Delft, The Netherlands

Current embedding algorithms' main limitation

Current methods rely on *heuristics* and use *likelihood optimization*.

These approaches yield fast and good results, but they provide little insight on the *likelihood's landscape*.

scientific reports

Check for updates

OPEN

Optimisation of the coalescent hyperbolic embedding of complex networks

Blanka Kovács¹ & Gergely Palla^{1,2}

Several observations indicate the existence of a latent hyperbolic space behind real networks that

New Journal of Physics

The open access journal at the forefront of physics

Deutscher Physikalischer Vereinigung DPG
IOP Institute of Physics

Published in partnership
with: Deutsche Physikalische
Gesellschaft and the Institute
of Physics

PAPER

Mercator: uncovering faithful hyperbolic embeddings of complex networks

Guillermo García-Pérez^{1,2,3}, Antoine Allard^{3,4,5}, M Ángeles Serrano^{3,4,5} and Marián Boguñá^{3,6}

¹ QTF Centre of Excellence, Turku Centre for Quantum Physics, Department of Physics and Astronomy, University of Turku, FI-20014 Turun Yliopisto, Finland

² Complex Systems Research Group, Department of Mathematics and Statistics, University of Turku, FI-20014 Turun Yliopisto, Finland

³ Department of Physics, University of Turku, FI-20014 Turun Yliopisto, Finland

198

IEEE/ACM TRANSACTIONS ON NETWORKING, VOL. 23, NO. 1, FEBRUARY 2015

Network Mapping by Replaying Hyperbolic Growth

Fragkiskos Papadopoulos, Constantinos Psomas, and Dmitri Krioukov

Abstract—Recent years have shown a promising progress in understanding geometric underpinnings behind the structure, function, and dynamics of many complex networks in nature and society. However, these promises cannot be readily fulfilled and lead to important practical applications, without a simple, reliable, and fast network mapping method to infer the latent geometric coordinates of nodes in a real network. Here, we present *HyperMap*, a simple method to map a given real network to its hyperbolic space. The method utilizes a recent geometric theory

complex networks [2]–[4].¹ A particular goal is to understand how these characteristics affect the various processes that run on top of these networks, such as routing, information sharing, data distribution, searching, and epidemics [2], [3], [5]. Understanding the mechanisms that shape the structure and drive the evolution of real networks can also have important applications in designing more efficient recommender and collaborative filtering systems [6] and for predicting missing and future

Current embedding algorithms' main limitation

Current methods rely on *heuristics* and use *likelihood optimization*.

These approaches yield fast and good results, but they provide little insight on the *likelihood's landscape*.

Goal: characterize the *embeddings' landscape*.

scientific reports

Check for updates

OPEN

Optimisation of the coalescent hyperbolic embedding of complex networks

Blanka Kovács¹ & Gergely Palla^{1,2,✉}

Several observations indicate the existence of a latent hyperbolic space behind real networks that

New Journal of Physics

The open access journal at the forefront of physics

Deutsche Physikalische Gesellschaft DPG
IOP Institute of Physics

Published in partnership with: Deutsche Physikalische Gesellschaft and the Institute of Physics

PAPER

Mercator: uncovering faithful hyperbolic embeddings of complex networks

Guillermo García-Pérez^{1,2,✉}, Antoine Allard^{1,4,5}, M Ángeles Serrano^{1,4,5} and Marián Boguñá^{1,6}

¹ QTF Centre of Excellence, Turku Centre for Quantum Physics, Department of Physics and Astronomy, University of Turku, FI-20014 Turun Yliopisto, Finland

² Complex Systems Research Group, Department of Mathematics and Statistics, University of Turku, FI-20014 Turun Yliopisto, Finland

198

IEEE/ACM TRANSACTIONS ON NETWORKING, VOL. 23, NO. 1, FEBRUARY 2015

Network Mapping by Replaying Hyperbolic Growth

Fragkiskos Papadopoulos, Constantinos Psomas, and Dmitri Krioukov

Abstract—Recent years have shown a growing interest in hyperbolic geometry as a powerful tool to understand the structure and dynamics of complex networks in nature and society. However, these processes cannot be readily fulfilled and lead to important practical applications, without a simple, reliable, and fast network mapping method to infer the latent geometric coordinates of nodes in a real network. Here, we present *HyperMap*, a simple method to map a given real network to its hyperbolic space. The method utilizes a recent geometric theory

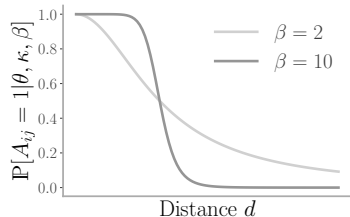
coalescent hyperbolic growth [1]. A natural goal is to understand the structure and dynamics of complex networks in nature and society. However, these processes cannot be readily fulfilled and lead to important practical applications, without a simple, reliable, and fast network mapping method to infer the latent geometric coordinates of nodes in a real network. Here, we present *HyperMap*, a simple method to map a given real network to its hyperbolic space. The method utilizes a recent geometric theory

on top of these networks, such as routing, information sharing, data distribution, searching, and epidemics [2], [3], [5]. Understanding the mechanisms that shape the structure and drive the evolution of real networks can also have important applications in designing more efficient recommender and collaborative filtering systems [6] and for predicting missing and future

$\mathbb{S}^1$ model

The likelihood of the $\mathbb{S}^1$ model¹ is

$$\mathbb{P}[G = g \mid \theta, \kappa, \beta] = \prod_{i < j} \mathbb{P}[A_{ij} = a_{ij} \mid \theta, \kappa, \beta],$$



1. Closely related to the $\mathbb{H}^2$ model which directly uses hyperbolic geometry (Krioukov, 2010).

$\mathbb{S}^1$ model

The likelihood of the $\mathbb{S}^1$ model¹ is

$$\mathbb{P}[G = g \mid \theta, \kappa, \beta] = \prod_{i < j} \mathbb{P}[A_{ij} = a_{ij} \mid \theta, \kappa, \beta],$$

where

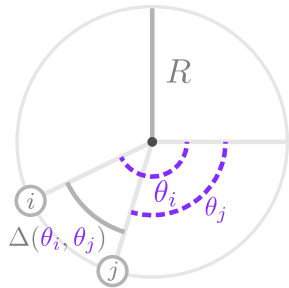
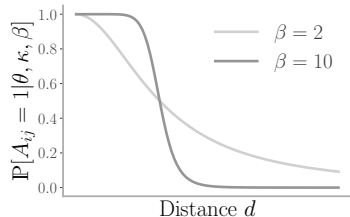
$$\mathbb{P}[A_{ij} = 1 \mid \theta, \kappa, \beta] = \frac{1}{1 + \left(\frac{R \Delta(\theta_i, \theta_j)}{\mu \kappa_i \kappa_j} \right)^\beta},$$

and

κ_i : expected degree of vertex i ;

β : controls the sharpness of the sigmoid;

$\Delta(\cdot, \cdot)$: angular separation.



1. Closely related to the $\mathbb{H}^2$ model which directly uses hyperbolic geometry (Krioukov, 2010).

$\mathbb{S}^1$ model

The likelihood of the $\mathbb{S}^1$ model¹ is

$$\mathbb{P}[G = g \mid \theta, \kappa, \beta] = \prod_{i < j} \mathbb{P}[A_{ij} = a_{ij} \mid \theta, \kappa, \beta],$$

where

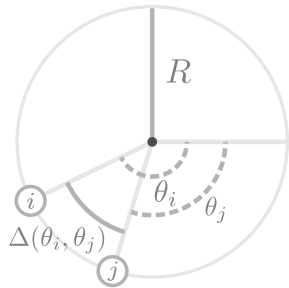
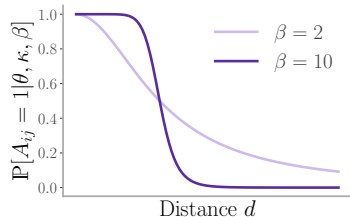
$$\mathbb{P}[A_{ij} = 1 \mid \theta, \kappa, \beta] = \frac{1}{1 + \left(\frac{R\Delta(\theta_i, \theta_j)}{\mu\kappa_i\kappa_j} \right)^\beta},$$

and

κ_i : expected degree of vertex i ;

β : controls the sharpness of the sigmoid;

$\Delta(\cdot, \cdot)$: angular separation.

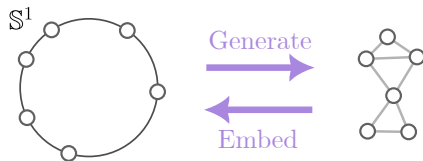


1. Closely related to the $\mathbb{H}^2$ model which directly uses hyperbolic geometry (Krioukov, 2010).

Embedding inference scheme

We infer the coordinates that generated a synthetic graph.

1. Choose the $\mathbb{S}^1$ model's parameters θ , κ and β .
2. Generate a synthetic graph with the likelihood $g \sim \mathbb{P}[G = g|\theta, \kappa, \beta]$.



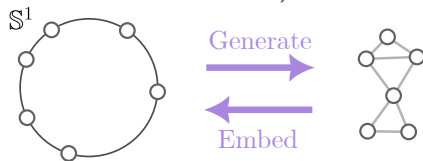
Embedding inference scheme

We infer the coordinates that generated a synthetic graph.

1. Choose the $\mathbb{S}^1$ model's parameters θ , κ and β .
2. Generate a synthetic graph with the likelihood $g \sim \mathbb{P}[G = g|\theta, \kappa, \beta]$.
3. Sample¹ the posterior obtained using Bayes' rule

$$f(\theta|G=g) = \frac{\pi(\theta)}{\mathbb{P}[G=g]} \prod_{i < j} \left(1 + \left(\frac{R\Delta(\theta_i, \theta_j)}{\mu\kappa_i\kappa_j} \right)^{\beta(2a_{ij}-1)} \right)^{-1},$$

$$\pi(\theta) = \left(\frac{1}{2\pi} \right)^n.$$



Issues with out-of-the-box HMC

Hamiltonian Monte Carlo (HMC) has poor mixing because of the *multimodality of the posterior*.

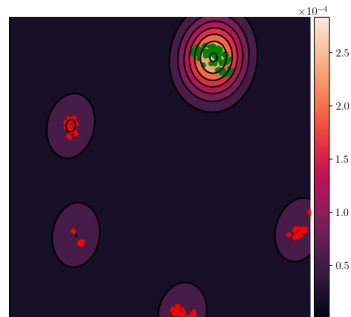


Figure: Multidimensional scaling (MDS) of the posterior sample obtained from 4 chains initialized randomly (red) and initialized at the ground truth (green).

Issues with out-of-the-box HMC

Hamiltonian Monte Carlo (HMC) has poor mixing because of the *multimodality of the posterior*.

Solved issues:

1. incorrect boundary periodicity;
2. symmetry-equivalent embeddings;
3. incoherent cluster alignments.

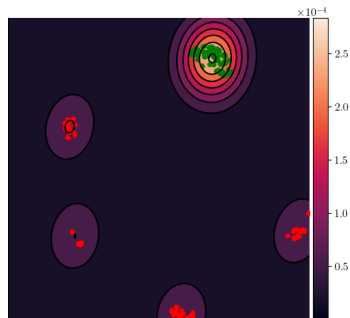


Figure: Multidimensional scaling (MDS) of the posterior sample obtained from 4 chains initialized randomly (red) and initialized at the ground truth (green).

Issues with out-of-the-box HMC

Hamiltonian Monte Carlo (HMC) has poor mixing because of the *multimodality of the posterior*.

Solved issues:

1. incorrect boundary periodicity;
2. symmetry-equivalent embeddings;
3. incoherent cluster alignments.

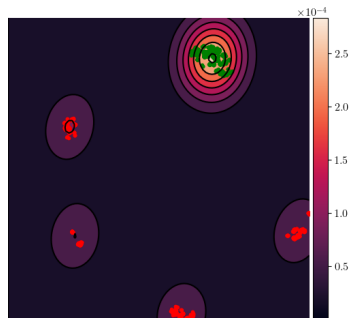
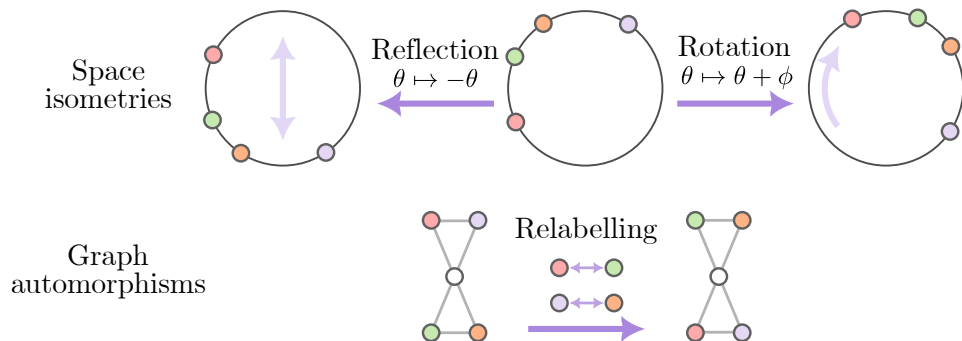
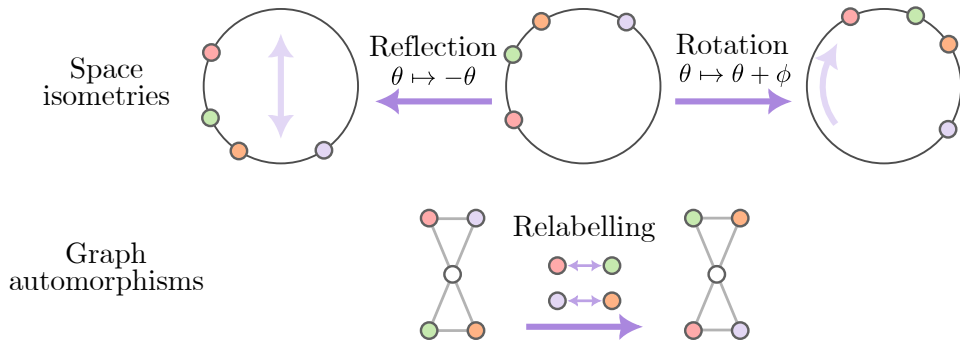


Figure: Multidimensional scaling (MDS) of the posterior sample obtained from 4 chains initialized randomly (red) and initialized at the ground truth (green).

Issue #2: Symmetries cause equivalent embeddings



Issue #2: Symmetries cause equivalent embeddings



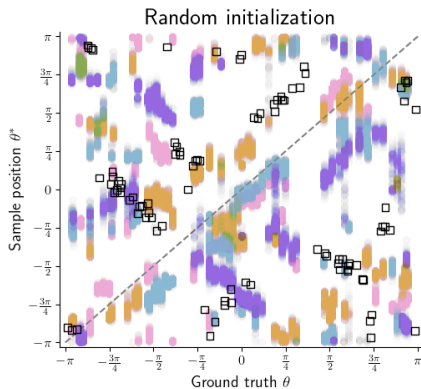
We *fix the vertex* of the largest degree at $\theta = 0$ to limit rotations. Samples are aligned² after running HMC.

2. Rotation ϕ which minimizes $\sum_i \Delta(\theta_i^* + \phi, \theta_i)^2$ across all combinations of automorphisms and reflections.

Issue #3: Clusters have different alignments

Superposition of the samples obtained from 4 chains. Each chain has a different color. Maximum *a posteriori* (MAP) shown with $\square$.

A *straight line* signals a perfect inference.

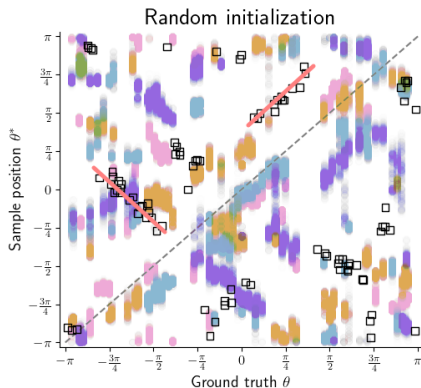


Issue #3: Clusters have different alignments

Superposition of the samples obtained from 4 chains. Each chain has a different color. Maximum *a posteriori* (MAP) shown with $\square$.

A *straight line* signals a perfect inference.

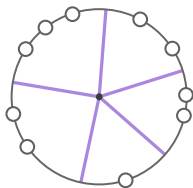
Issue: Clusters have *different alignments*.



Fixing cluster alignments with a new MCMC move

A *cluster angle swap* move helps the HMC sampler exit some local maxima.

1. Cluster identification

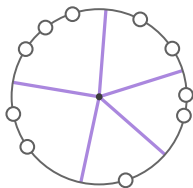


This move is its own inverse. Its acceptance probability depends only on the posterior ratio.

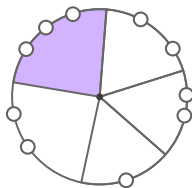
Fixing cluster alignments with a new MCMC move

A *cluster angle swap* move helps the HMC sampler exit some local maxima.

1. Cluster identification



2. Select cluster

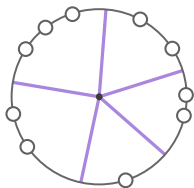


This move is its own inverse. Its acceptance probability depends only on the posterior ratio.

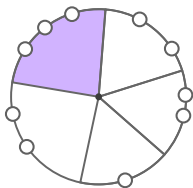
Fixing cluster alignments with a new MCMC move

A *cluster angle swap* move helps the HMC sampler exit some local maxima.

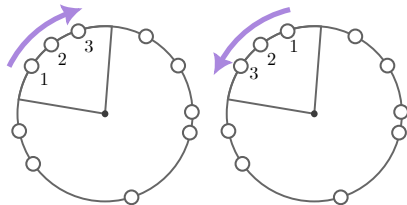
1. Cluster identification



2. Select cluster



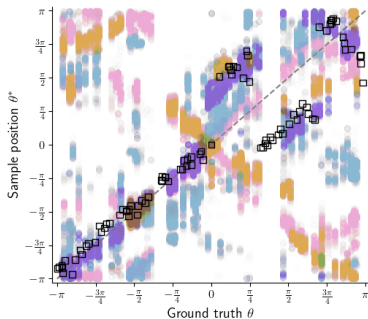
3. Reverse angles in cluster



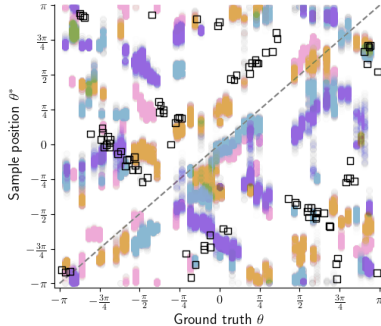
This move is its own inverse. Its acceptance probability depends only on the posterior ratio.

Sampling with HMC and cluster angle swap

With cluster angle swaps



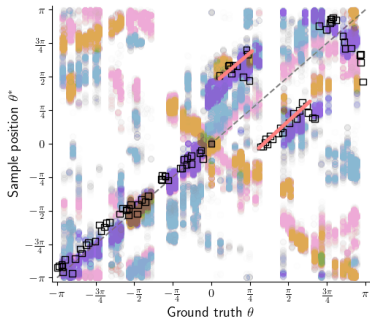
Only HMC



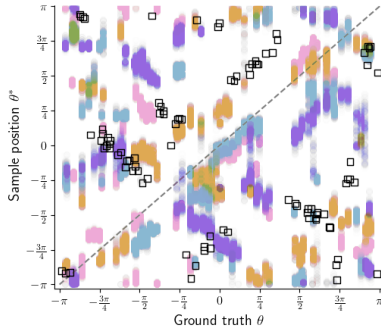
Sampling with HMC and cluster angle swap

Issue #4: clusters have incorrect relative positions.

With cluster angle swaps



Only HMC



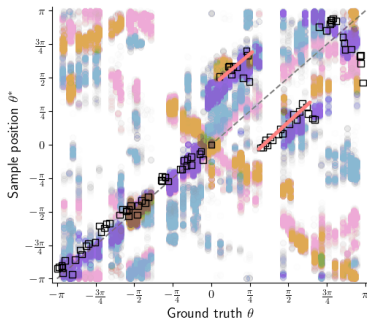
Sampling with HMC and cluster angle swap

Issue #4: clusters have incorrect relative positions.

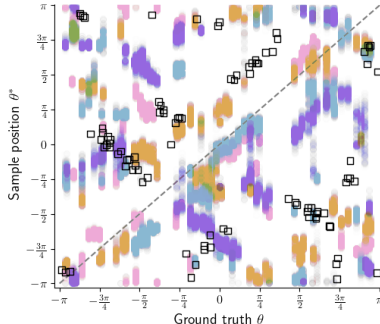
However, we believe these *clusters translations* could be, for certain graphs, nearly *equally probable*.

To be continued. . .

With cluster angle swaps



Only HMC



Summary

Next we want to

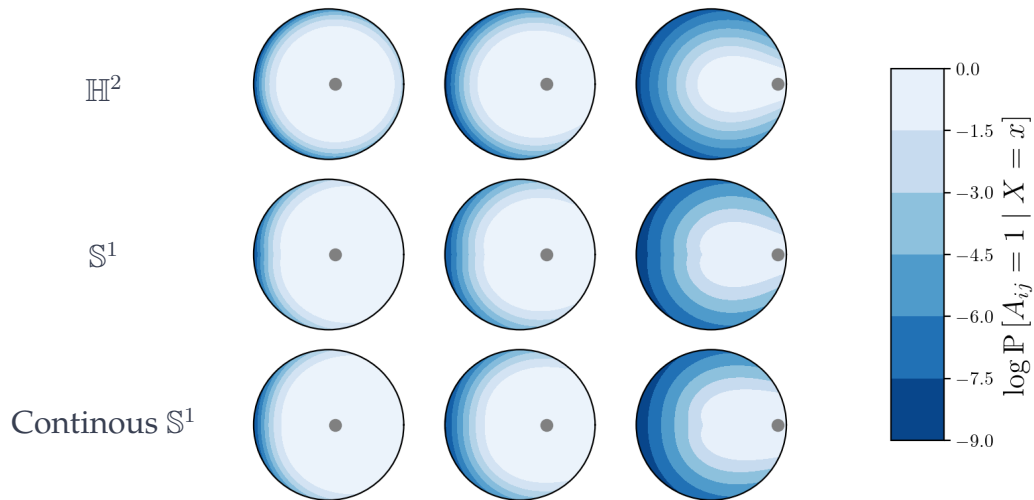
- improve the MCMC mixing such that the ground truth is accessible from any initialization;
- evaluate confidence intervals;
- infer the expected degrees κ ;
- compare our method to other algorithms on a large variety of graphs.

Takeaways:

- Our Bayesian approach can *characterize the embeddings' landscape*.
- The posterior is *multimodal* and thus difficult to explore with MCMC.
- There could be *multiple good embeddings*.

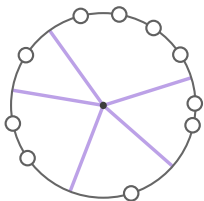
Feel free to contact me at ariane.lizotte.2@ulaval.ca.

$\mathbb{H}^2$ model vs $\mathbb{S}^1$ model: connection probability

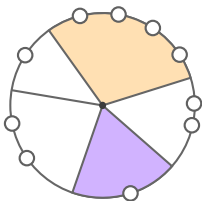


New move considered: cluster swapping

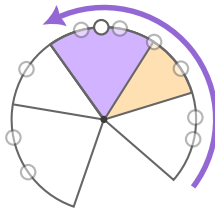
1. Cluster identification



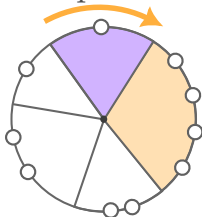
2. Select two clusters



3. Translate first cluster



4. Fill empty space



Sigmoid approximation of the absolute value

The angular separation $\Delta\theta_{ij}$ is not differentiable at every point

$$\Delta\theta_{ij} = \pi - |\pi - |\theta_i - \theta_j||.$$

The absolute value can be expressed with the Heaviside step function H

$$|x| = x(2H(x) - 1).$$

The step function is approximated with the sigmoid function σ_b

$$H(x) = \lim_{b \rightarrow \infty} \sigma_b(x)$$

$$\sigma_b(x) = \frac{1}{1 + e^{-bx}}.$$

